

Soft Spaces in Low-Qubit Hilbert Spaces – Phase 3.2

Whitepaper for the Algebraic Mechanism Programme:
 $\pm E$ Symmetry, Projector Dynamics, and the Route Toward an Effective Open-System Interpretation

Tom Stevns¹

¹Independent Researcher, Denmark

Research planning whitepaper – September 2026

Abstract

Phase 3.1 replaced basis-dependent eigenvector-pair hotspot scores by a basis-invariant projector formulation and isolated a small but reproducible REAL-specific enhancement of the otherwise largely structural $\pm E$ organization. The effect was independently replicated at 8Q and confirmed at 7Q and 9Q under predeclared acceptance criteria. Phase 3.2 therefore changes the scientific question. The primary task is no longer to establish whether a REAL-specific residual exists, but to identify the algebraic mechanism that produces the near-perfect $\pm E$ organization and to determine how that mechanism propagates through the effective projector response.

This whitepaper defines a falsifiable Phase-3.2 programme. The first target is an exact or approximate anticommuting symmetry operator S satisfying

$$SHS^{-1} \simeq -H, \quad \text{equivalently} \quad \{S, H\} \simeq 0,$$

which would imply paired energies $E \leftrightarrow -E$. The second target is to determine which Pauli terms preserve or violate this symmetry, and whether the REAL and NULL constructions transform differently under it. The third target is to propagate the symmetry through the Phase-3 effective operator

$$K_P^{(\eta)}(E) = PVQ \frac{E - QHQ}{(E - QHQ)^2 + \eta^2} QVP,$$

thereby asking whether the observed REAL-specific excess can be expressed as a symmetry-sensitive interference or cancellation term. Only after these algebraic tests succeed should the P/Q decomposition be promoted toward a Feshbach, self-energy, or open-quantum-system interpretation.

The programme is organized as a sequence of versioned experiments beginning with v32_0. Each experiment has explicit invariants, null hypotheses, acceptance gates, and stop conditions. The intended outcome is not merely a larger numerical catalogue, but a compact mechanism statement capable of explaining why REAL remains close to perfect $\pm E$ organization while NULL approaches it with increasing Hilbert-space dimension.

Status: planning document. No Phase-3.2 result is claimed in this whitepaper.

Repository convention: Phase 3.1 source/output files are named v31_0–v31_8. Historical frozen computational namespaces such as v30.2 remain unchanged where they are part of seed generation or provenance. Phase 3.2 begins at v32_0.

Contents

1 Starting point: what Phase 3.1 has established

2

2	Primary Phase-3.2 hypothesis: an anticommuting symmetry	2
2.1	Exact symmetry	2
2.2	Approximate symmetry	3
3	Why Pauli algebra makes the hypothesis testable	3
4	v32_0: exact symmetry audit	4
4.1	Purpose	4
4.2	Required outputs per seed	4
4.3	Predeclared interpretation	4
5	v32_1: term-level symmetry anatomy	5
6	v32_2: symmetry of the perturbations and REAL/NULL distinction	5
7	v32_3: propagation into the effective projector operator	6
8	A candidate Phase-3.2 mechanism equation	6
9	v32_4: controlled symmetry breaking	7
10	v32_5: dimension scaling as a derived, not fitted, test	7
11	Relation to Feshbach projection and open quantum systems	8
12	Predeclared statistical discipline	8
13	Proposed version map	9
14	Stop conditions and falsification criteria	9
15	What Phase 3.2 should be able to explain	9
16	Working physical picture	10
17	Immediate next action	10
18	Conclusion	11

1 Starting point: what Phase 3.1 has established

Phase 3.1 produced three methodological conclusions that should be treated as fixed input rather than reopened by default.

First, arbitrary selected eigenvectors inside an exactly degenerate eigenspace are not legitimate physical observables. If the score changes under a unitary rotation within a degenerate subspace while the Hamiltonian remains unchanged, the score is representation dependent. The Phase-3 response was therefore reformulated in terms of complete eigenspace projectors.

Second, the relevant perturbative object is an operator on a complete target subspace $P\mathcal{H}$, not a scalar attached to an arbitrary pair. For $Q = \mathbb{I} - P$, perturbation V , reference energy E , and regularization $\eta > 0$, the working operator is

$$K_P^{(\eta)}(E) = PVQ \frac{E - QHQ}{(E - QHQ)^2 + \eta^2} QVP. \quad (1)$$

Third, after REAL/NULL decomposition, the dominant $\pm E$ correlation is shared by both ensembles, but REAL retains a smaller positive excess. The Phase-3.1 endpoint was

$$\Delta r(Q) = r_{\text{REAL}}(Q) - r_{\text{NULL}}(Q), \quad (2)$$

with independent 8Q replication and cross-dimensional confirmation at 7Q and 9Q.

The observed dimensional pattern motivates Phase 3.2:

$$r_{\text{REAL}}(Q) \approx 1, \quad r_{\text{NULL}}(Q) \uparrow 1, \quad \Delta r(Q) \downarrow \quad \text{for } Q = 7, 8, 9. \quad (3)$$

Equation (3) is an empirical pattern, not yet a scaling law. The immediate question is therefore mechanistic: what algebraic structure makes the spectrum and response so strongly paired under $E \mapsto -E$?

2 Primary Phase-3.2 hypothesis: an anticommuting symmetry

2.1 Exact symmetry

Suppose a unitary or antiunitary operator S exists such that

$$SHS^{-1} = -H. \quad (4)$$

For unitary S , this is equivalent to

$$\{S, H\} = SH + HS = 0. \quad (5)$$

If $H|\psi_E\rangle = E|\psi_E\rangle$, then

$$HS|\psi_E\rangle = -SH|\psi_E\rangle \quad (6)$$

$$= -ES|\psi_E\rangle, \quad (7)$$

so every nonzero energy eigenstate has a partner at $-E$. Thus exact anticommutation gives a direct algebraic explanation of spectral $\pm E$ pairing.

The strongest form of the hypothesis is therefore:

The random-Pauli Hamiltonian ensemble used in Phase 3.1 possesses, seed by seed or with high probability, an exact Pauli-product symmetry S that anticommutes with all active Hamiltonian terms.

2.2 Approximate symmetry

Exact anticommutation is not required for a useful mechanism. Define the normalized anticommutator defect

$$\epsilon_S(H) = \frac{\|SH + HS\|_F}{2\|H\|_F}. \quad (8)$$

Then $\epsilon_S = 0$ is exact symmetry, while $0 < \epsilon_S \ll 1$ defines an approximate chiral-like symmetry. A Phase-3.2 symmetry audit should report the minimum

$$\epsilon_*(H) = \min_{S \in \mathcal{C}} \epsilon_S(H), \quad (9)$$

where $\mathcal{C}$ is a frozen candidate class, initially the n -qubit Pauli group modulo irrelevant global phases.

A complementary spectral diagnostic is

$$\epsilon_{\text{spec}}(H) = \frac{1}{N} \sum_{j=1}^N \frac{|E_j + E_{N+1-j}|}{E_{\text{scale}} + \varepsilon}, \quad (10)$$

for sorted eigenvalues $E_1 \leq \dots \leq E_N$. Algebraic and spectral defects should be studied together; agreement between them is much stronger evidence than either alone.

3 Why Pauli algebra makes the hypothesis testable

Let the Hamiltonian be

$$H = \sum_{a=1}^m h_a P_a, \quad (11)$$

where each P_a is an n -qubit Pauli string. For a candidate Pauli string S , each term either commutes or anticommutes with S :

$$SP_a = \chi_a P_a S, \quad \chi_a \in \{+1, -1\}. \quad (12)$$

Consequently,

$$\{S, H\} = \sum_a h_a (1 + \chi_a) P_a S. \quad (13)$$

If S anticommutes with every active P_a , then $\chi_a = -1$ for all a and $\{S, H\} = 0$ exactly.

This turns the first Phase-3.2 experiment into a finite binary-algebra problem. A Pauli string can be represented by a binary symplectic vector

$$p = (\mathbf{x} \mid \mathbf{z}) \in \mathbb{F}_2^{2n}. \quad (14)$$

Two Pauli strings p and s anticommute exactly when their symplectic inner product is one:

$$\langle s, p \rangle_{\text{sp}} = \mathbf{x}_s \cdot \mathbf{z}_p + \mathbf{z}_s \cdot \mathbf{x}_p \pmod{2} = 1. \quad (15)$$

Therefore, demanding that a single S anticommute with every Hamiltonian term yields the linear system

$$As = \mathbf{1} \pmod{2}, \quad (16)$$

where each row of A encodes the symplectic constraint generated by one active Pauli term.

Equation (16) is potentially the most important practical simplification in Phase 3.2. Instead of searching numerically over all 4^n operators, the existence and dimension of the exact Pauli-symmetry solution space can be obtained by Gaussian elimination over $\mathbb{F}_2$.

4 v32_0: exact symmetry audit

4.1 Purpose

v32_0 should answer one question only:

Does the frozen 11-term random-Pauli Hamiltonian possess an exact Pauli-product operator S that anticommutes with all active Hamiltonian terms?

The initial implementation should use the same Hamiltonian construction already validated in Phase 3.1. It should not modify the REAL/NULL construction, perturbation families, prominence definition, or replication statistics.

4.2 Required outputs per seed

For each seed and dimension, report:

1. rank of the binary constraint matrix A ;
2. consistency of $As = \mathbf{1}$ over $\mathbb{F}_2$;
3. dimension of the solution affine space if consistent;
4. one canonical nontrivial solution S_* ;
5. numerical anticommutator defect $\epsilon_{S_*}(H)$;
6. spectral pairing defect $\epsilon_{\text{spec}}(H)$;
7. maximum absolute difference between numerically paired eigenvalues $|E_j + E_{N+1-j}|$.

4.3 Predeclared interpretation

Three outcomes are possible.

Outcome	Observation	Interpretation
A	Exact S found, ϵ_S at machine precision	Strong algebraic explanation of $\pm E$ spectral pairing. Proceed to symmetry propagation.
B	No exact S , but very small ϵ_* and spectral defect	Approximate/emergent symmetry. Identify symmetry-breaking terms.
C	No small-defect S and no stable relation to spectral pairing	Abandon the simple chiral-symmetry mechanism and broaden the operator class.

Table 1: Predeclared decision logic for v32_0.

A null result in Outcome C is scientifically useful: it prevents the project from retrofitting an attractive symmetry interpretation onto a numerical correlation.

5 v32_1: term-level symmetry anatomy

If v32_0 identifies an exact or approximate S , the next task is to determine which Hamiltonian terms enforce or violate it.

Partition the Hamiltonian into

$$H = H_- + H_+, \quad (17)$$

where

$$\{S, H_-\} = 0, \quad [S, H_+] = 0. \quad (18)$$

Then

$$\{S, H\} = 2SH_+, \quad (19)$$

so H_+ is precisely the symmetry-breaking component relative to S . A natural normalized breaking amplitude is

$$\beta_S = \frac{\|H_+\|_F}{\|H\|_F}. \quad (20)$$

The version should perform controlled leave-one-term-out and add-one-term perturbations. The causal expectation is:

$$\beta_S \uparrow \implies \epsilon_{\text{spec}} \uparrow, \quad r_{\pm E} \downarrow. \quad (21)$$

The relationship need not be linear, but it should be monotone enough to distinguish mechanism from coincidence.

A particularly strong result would be a prospective test: determine S and the preserving/breaking classification from the Pauli strings alone, before diagonalizing H , and then correctly predict which modified Hamiltonians retain or lose $\pm E$ pairing.

6 v32_2: symmetry of the perturbations and REAL/NULL distinction

The spectrum alone cannot explain the Phase-3.1 endpoint because NULL also becomes strongly $\pm E$ organized. The key question is how the perturbation operator V transforms under S .

Define a parity label σ_V when possible by

$$SVS^{-1} = \sigma_V V, \quad \sigma_V \in \{+1, -1\}. \quad (22)$$

More generally decompose

$$V = V_+ + V_-, \quad (23)$$

with

$$V_+ = \frac{1}{2}(V + SVS^{-1}), \quad V_- = \frac{1}{2}(V - SVS^{-1}). \quad (24)$$

Then V_+ is symmetry even and V_- is symmetry odd.

This decomposition should be computed separately for the dephasing family Z/ZZ and transverse family X/XX . The central Phase-3.2 question becomes whether the REAL-specific residual is associated with a reproducible imbalance in the way V_+ and V_- couple $+E$ and $-E$ subspaces.

For matched projectors P_E and P_{-E} , define channel strengths

$$M_{\alpha\beta}(E) = \|P_{\alpha E} V P_{\beta E}\|_F^2, \quad \alpha, \beta \in \{+, -\}. \quad (25)$$

Candidate symmetry-sensitive contrasts include

$$\Xi_V(E) = \frac{M_{+-}(E) - M_{++}(E)}{M_{+-}(E) + M_{++}(E) + \varepsilon} \quad (26)$$

and their REAL-minus-NULl differences.

The aim is not to search many contrasts and select the most favorable one. A small set should be frozen in advance from the symmetry algebra and tested unchanged across seeds and dimensions.

7 v32_3: propagation into the effective projector operator

Phase 3.1 established $K_P^{(\eta)}(E)$ as the basis-invariant second-order object. Phase 3.2 should now ask how K_P transforms under the candidate symmetry.

Let P_E project onto the exact eigenspace at energy E . Under exact symmetry,

$$SP_ES^{-1} = P_{-E}. \quad (27)$$

For the regularized odd kernel

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}, \quad g_\eta(-x) = -g_\eta(x), \quad (28)$$

one can compare

$$K_{P_{-E}}^{(\eta)}(-E) \quad \text{with} \quad SK_{P_E}^{(\eta)}(E)S^{-1}. \quad (29)$$

The transformation sign depends on the parity structure of V and on the precise P/Q mapping. Instead of assuming the answer, v32_3 should calculate the operator residual

$$\mathcal{R}_K(E; \tau) = \frac{\|K_{P_{-E}}^{(\eta)}(-E) - \tau SK_{P_E}^{(\eta)}(E)S^{-1}\|_F}{\|K_{P_{-E}}^{(\eta)}(-E)\|_F + \|K_{P_E}^{(\eta)}(E)\|_F + \varepsilon}, \quad (30)$$

for $\tau \in \{+1, -1\}$ and choose the sign by prior algebraic derivation, not by post-hoc minimization in the final test.

If a near-exact transformation law holds for REAL but is weakened by the NULL basis randomization, the Phase-3.1 endpoint Δr would acquire a direct operator-level explanation.

8 A candidate Phase-3.2 mechanism equation

The working conceptual chain is

$$\boxed{\{S, H\} \approx 0 \implies P_E \leftrightarrow P_{-E} \implies K_{P_E}(E) \leftrightarrow K_{P_{-E}}(-E) \implies r_{\text{REAL}} > r_{\text{NULL}}.} \quad (31)$$

A useful mechanism score should combine symmetry quality with the REAL-specific projector-response excess. One possible pre-theoretical form is

$$\boxed{\mathcal{M}_{\text{SS}} = (1 - \epsilon_*(H)) [r_{\text{REAL}} - r_{\text{NULL}}],} \quad (32)$$

where ϵ_* is normalized to $[0, 1]$ for the chosen candidate class. Equation (32) is not proposed as a final physical law. Its role is diagnostic: a genuine symmetry mechanism should cause the REAL-specific excess to weaken when the relevant symmetry is deliberately broken.

The decisive causal experiment is therefore not another passive correlation. It is an intervention:

$$H(\lambda) = H_- + \lambda H_+, \quad (33)$$

with λ varied on a frozen grid. If the hypothesis is correct, increasing $|\lambda|$ should degrade spectral pairing and, in a predictable way, degrade the REAL-specific projector organization.

9 v32_4: controlled symmetry breaking

v32_4 should be the first deliberately causal test in the programme. Once S is fixed from v32_0, construct H_- and H_+ as above and evaluate

$$H(\lambda) = H_- + \lambda H_+, \quad (34)$$

for a predeclared grid, for example

$$\lambda \in \{0, 0.25, 0.5, 1, 2\}.$$

The exact grid should be frozen before observing the endpoint.

For each λ , calculate:

1. anticommutator defect ϵ_S ;
2. spectral pairing defect ϵ_{spec} ;
3. r_{REAL} and r_{NULL} ;
4. Δr ;
5. projector transformation residual $\mathcal{R}_K$.

A mechanism-level success requires coordinated movement, not merely a significant p -value. The predicted ordering is

$$\epsilon_S \uparrow, \quad \epsilon_{\text{spec}} \uparrow, \quad \mathcal{R}_K \uparrow, \quad \Delta r \text{ changes systematically according to the derived symmetry law.} \quad (35)$$

If Δr remains unchanged while the $\pm E$ symmetry is destroyed, then the symmetry may explain the paired spectrum but not the REAL-specific Soft-Space residual. That distinction is crucial.

10 v32_5: dimension scaling as a derived, not fitted, test

Only after a mechanism survives v32_0–v32_4 should dimension scaling be revisited. Phase 3.1 suggests

$$\Delta r(7) > \Delta r(8) > \Delta r(9) > 0. \quad (36)$$

The wrong next step would be to fit an arbitrary curve through three points. The right next step is to derive a quantity from the symmetry/null construction that is expected to scale with Hilbert-space dimension.

For example, if NULL randomization produces concentration of measure around a symmetry-compatible average, one might expect a residual variance of schematic form

$$\text{Var}_{\text{NULL}}(X_Q) \sim D^{-\alpha}, \quad D = 2^Q, \quad (37)$$

for some derived observable X_Q and exponent α determined by the model. This is presently only a theoretical possibility. The exponent must not be inferred from the three existing Δr points and then declared predictive.

A publication-grade scaling test should use new, disjoint seed sets at each dimension. This also removes the mild cross-dimensional dependence introduced when the same seed identifiers were used for 7Q and 9Q in the dimensional replication.

11 Relation to Feshbach projection and open quantum systems

The P/Q decomposition in Eq. (1) is mathematically close to projection-operator methods. In the standard Feshbach construction, eliminating the Q sector gives an energy-dependent effective operator on $P\mathcal{H}$,

$$H_{\text{eff}}(E) = PHP + PHQ(E - QHQ)^{-1}QHP, \quad (38)$$

up to conventions and the particular partition of the interaction [1, 2]. The Phase-3 operator is structurally analogous, with a regularized resolvent and perturbation V .

Open-quantum-system theory adds a stronger physical interpretation. There one identifies a system sector and environmental degrees of freedom and studies a reduced density operator such as

$$\rho_P(t) = \text{Tr}_Q \rho(t), \quad (39)$$

with memory kernels, dissipation, decoherence, or effective non-Hermitian self-energies [3, 4, 5].

Phase 3.2 should *not* assume that the present Q space is a physical bath. Three milestones are required before that interpretation becomes credible:

1. the algebraic $\pm E$ mechanism must be identified;
2. its transformation through $K_P(E)$ must be derived and numerically verified;
3. an operational system/environment partition or reduced-dynamics observable must be introduced.

If these milestones are met, a later Phase 3.3 or Phase 4 could test whether Soft Spaces correspond to subspaces with anomalously weak effective leakage, suppressed decoherence, or destructive interference of virtual transitions through Q .

12 Predeclared statistical discipline

Phase 3.2 is mechanism-driven, but statistical discipline remains necessary. The following rules should be frozen before each new replication block:

1. Separate development seeds from confirmation seeds.
2. Use disjoint seed sets across dimensions for scaling claims.
3. Freeze candidate operator class $\mathcal{C}$ before inspecting endpoint behavior.
4. Freeze thresholds for “exact”, “approximate”, and “broken” symmetry before final runs.
5. Report all seeds, including failures and singular/degenerate cases.
6. Preserve raw output files unchanged as computational provenance.
7. Preserve frozen RNG namespace strings even when repository filenames are refactored.
8. Do not tune η , neighborhood radius, Pauli term count, or perturbation families after seeing a target result unless the run is explicitly reclassified as development-only.

For binary seedwise endpoints, exact sign tests remain useful. For continuous paired endpoints, Wilcoxon signed-rank tests and bootstrap intervals remain appropriate companions. However, a Phase-3.2 mechanism claim should be based primarily on prederived operator identities and successful intervention tests, not on significance alone.

13 Proposed version map

Version	Primary task	Main deliverable
v32_0	Exact symmetry audit	Solve $As = \mathbf{1}$ over $\mathbb{F}_2$; verify $\{S, H\} = 0$ numerically; quantify spectral pairing.
v32_1	Term anatomy	Identify symmetry-preserving and symmetry-breaking Pauli terms; prospective leave-one/add-one predictions.
v32_2	Perturbation symmetry	Decompose $V = V_+ + V_-$ and test symmetry-sensitive REAL/NULL channel contrasts.
v32_3	Projector propagation	Derive and test the $E \leftrightarrow -E$ transformation law of $K_P^{(\eta)}(E)$.
v32_4	Causal symmetry breaking	Intervene with $H(\lambda) = H_- + \lambda H_+$ and test coordinated degradation of symmetry and response.
v32_5	Dimensional mechanism test	Use new disjoint seeds and a derived scaling prediction rather than a post-hoc fit.

Table 2: Recommended Phase-3.2 implementation sequence.

The version map is deliberately conservative. A later version should not be started merely because the previous script ran. Each step should either establish the required mechanism ingredient or explicitly falsify it.

14 Stop conditions and falsification criteria

A strong research programme needs conditions under which a favored explanation is rejected.

The simple anticommuting-symmetry hypothesis should be abandoned or substantially revised if any of the following occurs:

1. no exact or low-defect S exists in a well-motivated operator class;
2. spectral $\pm E$ pairing remains strong when the candidate symmetry is deliberately destroyed;
3. the transformation law of K_P fails even when the spectral symmetry is exact;
4. REAL/NULL differences remain statistically unchanged under controlled symmetry breaking;
5. the apparent mechanism depends on post-hoc choice of S , perturbation family, regularization, or seed subset;
6. independent seeds fail to reproduce the mechanism-level endpoint.

Conversely, the mechanism hypothesis would become substantially stronger if a candidate S can be derived from the Pauli term set before diagonalization, predicts the spectrum, predicts the projector transformation law, and predicts how deliberate symmetry breaking changes Δr .

15 What Phase 3.2 should be able to explain

A successful Phase-3.2 mechanism should explain, in one coherent framework, at least the following observations from Phase 3.1:

1. why $+E$ and $-E$ eigenspaces are paired so strongly;

2. why the pairing is present in NULL as well as REAL;
3. why REAL is systematically closer to perfect pairing;
4. why the REAL-specific excess survives when exact degeneracy disappears;
5. why the REAL–NULL gap decreases strongly from 7Q to 9Q;
6. why a complete-projector formulation is necessary for basis invariance.

The fourth point is particularly important. A 7Q case with completely nondegenerate eigenspaces nevertheless exhibited a large positive REAL–NULL correlation difference in Phase 3.1. Therefore degeneracy may amplify methodological problems, but it cannot be the sole generator of the new REAL-specific effect.

16 Working physical picture

The most economical current picture is the following.

The Hamiltonian may possess a chiral-like algebraic symmetry that organizes its spectrum into $\pm E$ partners. This symmetry is largely responsible for the high correlation seen in both REAL and NULL. The physical perturbations then probe complete spectral subspaces through second-order virtual transitions mediated by Q . REAL retains coherent geometric information about how those subspaces and perturbations are aligned, whereas the spectrum-matched NULL construction destroys part of that alignment while preserving much of the underlying spectral symmetry. The measured Phase-3.1 quantity Δr is therefore interpreted not as the symmetry itself, but as a residual organization above a symmetry-dominated background.

In schematic form,

$$\boxed{\text{Pauli algebra} \rightarrow \pm E \text{ spectral symmetry} \rightarrow \text{projector-paired virtual channels} \rightarrow \text{REAL-specific coherence/cancellation}} \quad (40)$$

Every arrow in Eq. (40) is a testable statement. Phase 3.2 should be considered complete only when the central arrows have either been demonstrated or rejected.

17 Immediate next action

The recommended next action is to implement `v32_0` and nothing broader. It should reuse the frozen Phase-3.1 Hamiltonian generator, extract the active Pauli strings, construct the binary symplectic constraint system in Eq. (16), solve it exactly over $\mathbb{F}_2$, and verify the resulting candidate operator directly in matrix form.

A minimal successful `v32_0` report should fit on one screen per seed and end with a compact summary such as:

```
seed      Q    rank(A)  exact_S  dim(solution)  eps_anticom  eps_spectrum
...
```

Decision:

```
EXACT ANTICOMMUTING PAULI SYMMETRY FOUND / NOT FOUND
```

No REAL/NULL endpoint should be optimized in `v32_0`. That separation is deliberate: first establish the symmetry of H , then ask how the perturbation response sees it.

18 Conclusion

Phase 3.2 should be a mechanism phase. The existing evidence is already sufficient to justify moving beyond repeated hotspot discovery. The highest-value next question is whether the strong $\pm E$ organization has an exact algebraic origin in the Pauli Hamiltonian ensemble.

The proposed programme begins with an anticommutation audit, proceeds through term-level and perturbation-level symmetry analysis, then propagates the result into the basis-invariant projector operator, and finally tests causality by controlled symmetry breaking. Only after those steps should the project attempt a genuine open-system interpretation or a dimensional scaling law.

The central standard for success is therefore stronger than “the numbers remain significant.” A successful Phase 3.2 should produce a compact derivation that predicts the observed structure and survives prospective intervention. If it does, Soft Spaces will have moved from a reproducible numerical phenomenon toward an identifiable algebraic mechanism.

References

- [1] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* **5**, 357–390 (1958).
- [2] H. Feshbach, “A unified theory of nuclear reactions. II,” *Annals of Physics* **19**, 287–313 (1962).
- [3] S. Nakajima, “On quantum theory of transport phenomena: steady diffusion,” *Progress of Theoretical Physics* **20**, 948–959 (1958).
- [4] R. Zwanzig, “Ensemble method in the theory of irreversibility,” *Journal of Chemical Physics* **33**, 1338–1341 (1960).
- [5] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002).
- [6] A. Altland and M. R. Zirnbauer, “Nonstandard symmetry classes in mesoscopic normal-superconducting hybrid structures,” *Physical Review B* **55**, 1142–1161 (1997).